Coursework Assessment 1 Time Domain(Select one or more)

Signal and System Time Domain Analysis MATLAB Simulation

Topic 1: Basic Signal Generation, Voice Reading and Playback

1) Generate a sinusoidal signal, change the angular frequency and initial phase of the sinusoidal signal, observe the waveform change;
2) Generate a square wave with an amplitude of 1, a fundamental frequency of $2Hz$, and a duty cycle of 50%.
3) Observe changes in the stock index during a certain period of time, and generate index signals that simulate changes.
4) Record a piece of audio signal to read and play the audio signal.

Topic 2: The basic operation of the signal (inversion, scaling of speech signals)

1) The original audio signal is extended and compressed in the time domain.
2) Amplify and reduce the amplitude of the original audio signal in the frequency domain.
3) Invert the original audio signal in the time domain.

Topic 3: System Response Time Domain Solution

1) Find the zero-input response and zero-state response of an RLC circuit;
2) Mix the noise into the original audio signal, then use the M-point moving average system to denoise the noise-interfered signal, change the number of M points, and compare the denoising effect under different points.

Topic 4: Approximate calculation of continuous signal convolution

The convolution of two continuos signals is defined as

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$$

For numerical calculations, continuous signals need to be sampled. Record $x[k]=x(k\Delta)$, $h[k]=h(k\Delta)$, and Δ is the sampling interval for numerical calculation. The continuous signal convolution can be written approximately as $y(k\Delta) \approx \Delta x[k] * h[k]$.

This can use the *conv* function to approximate the convolution of continuous signals. Let $x(t)=u(t)-u(t-1)$, $h(t)=x(t)*x(t)$.

(a) In order to compare with the results of the approximate calculation, use analytics to find $y(t)=x(t)*h(t)$;

(b) Calculate the approximate numerical value of the convolution using different Δ and compare it with the result in (a);

(c) Prove that formula (1) is established;

(d) If *x(t)* and *h(t)* are not time-limited signals, what are the problems encountered in the approximate calculation using the above method? Give a solution to the problem;

(e) If *x(t)* and *h(t)* are approximated as

$$x(t) \approx \sum_{n=-\infty}^{\infty} x(n\Delta)p_{\Delta}(t-0.5\Delta-n\Delta)$$

$$h(t) \approx \sum_{n=-\infty}^{\infty} h(n\Delta) p_{\Delta}(t - 0.5\Delta - n\Delta)$$

Derive an approximate algorithm for calculating convolutions. Take the same sampling interval and compare the calculated convolution error of the two methods.